

Applied Statistics for Engineers and Scientists II - 23TT2

Lab 03: Ridge and Lasso regression

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Problem 1: Ridge and LASSO regression

Download the dataset `dummy.csv` in Lab 02.

- (a) Randomly split the dataset into two subsets, called `train.data` and `test.data` (recall the function at the end of Lab 02).
- (b) Consider the 5th-order polynomial regression model:

$$y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \beta_4 X^4 + \beta_5 X^5 + \epsilon, \epsilon \sim \mathcal{N}(0, \sigma^2)$$

Estimate the coefficients of this model based on the training dataset. Compute the MSE on the testing dataset. Comment on the result.

- (c) On the training dataset, perform cross validation of the ridge regression model. Which λ performs best on the training data? Compute the estimated coefficients with the optimal λ , and compare it to the OLS one.
- (d) Do the same with the Lasso regression model. Between ridge and lasso models, which one that has coefficients prefer towards the cubic model?

Problem 2: Abalone dataset

Download the dataset `abalone.csv` dataset: [Link](#).

- (a) Randomly split the dataset into two subsets, called `train.data` and `test.data`. Then, perform a full regression model:

$$\begin{aligned} \text{Whole_weight} = & \beta_0 + \beta_1 \text{Length} + \beta_2 \text{Diameter} + \beta_3 \text{Height} + \beta_4 \text{Shucked_weight} \\ & + \beta_5 \text{Viscera_weight} + \beta_6 \text{Shell_weight} + \epsilon \end{aligned}$$

Then, use the lasso for variable selection using cross validation. Which variables do you keep?

- (b) Install the package called `olsrr`. This package helps performing stepwise variable selection with various criterions.
The function `ols_step_all_possible` will fit all regressions involving all possible subsets of the set of potential independent variables.
In the model above, how many possible sub-models are there? Run this function, and plot the subset sizes vs the AIC scores. Interpret the plot and determine which variables should be kept.
Note. Alternatively, you can use the function `ols_step_best_subset` with parameter `metric = "aic"`
- (c) Read the documentation of the functions `ols_step_backward_aic` and `ols_step_forward_aic` by this link here: [Documentation](#).

Then, perform backward/forward stepwise selection methods, and comment on the result. Are the best subsets using these methods same as the previous ones ?

Problem 3: Prostate dataset

Download the `prostate.csv` dataset on the course website.

Dataset context: A study on 97 men with prostate cancer who were due to receive a radical prostatectomy. The `prostate` data frame has 97 rows and 9 columns:

- `lcavol` : Log-transformed cancer volume.
- `lweight` : Log-transformed prostate weight.
- `age` : Patient's age in years.
- `lbph` : Log-transformed amount of benign prostatic hyperplasia.
- `svi` : Presence of seminal vesicle invasion, a binary variable indicating whether the cancer has spread to the seminal vesicles.
- `lcp` : Log-transformed capsular penetration.
- `gleason` : Gleason score.
- `pgg45` : Percentage of Gleason scores 4 or 5, indicating the proportion of high-grade cancer patterns.
- `lpsa` : Log-transformed prostate-specific antigen (PSA) level.

- (a) Make a regression model predicting `lpsa` using every other variables. Compare this to the noise-only model using an appropriate statistical test:

$$\text{lpsa} = \beta_0 + \epsilon, \epsilon \sim \mathcal{N}(0, \sigma^2)$$

Comment on the result.

- (b) Given the grid search for λ is `exp(seq(-5, 5, by = 0.01))`, find the optimal λ for the ridge regression.
Which variables do you consider to leave out of the model ? Compute the MSE on the testing dataset.
- (c) Do the same with the lasso regression. With the optimal λ , which coefficients remain to be non-zero ? Compare the estimated coefficients and MSE on the testing dataset between ridge and lasso models.
- (d) Perform the forward/backward stepwise selection methods for this model. Compare the optimal choice of variables using this method with previous ones.
- (e) Find the best 3 variables to predict `lpsa` for this model. Here, "best" means the lowest AIC score.

Problem 4: Practice

Download the `fat.csv` dataset on the course website.

Dataset context:

- **brozek** – Percent body fat calculated using Brozek’s equation: $\frac{457}{\text{Body Density}} - 414.2$
- **siri** – Percent body fat calculated using Siri’s equation: $\frac{495}{\text{Body Density}} - 450$
- **density** : Body density in grams per cubic centimeter (gm/cm^3)
- **age** : Age in years
- **weight** : Weight in pounds (lbs)
- **height** : Height in inches
- **adipos** : Adiposity index, defined as Weight divided by Height squared (kg/m^2)
- **free** : Fat-free weight calculated as $(1 - \text{fraction of body fat}) \times \text{Weight}$ using Brozek’s formula
- **neck** : Neck circumference in centimeters
- **chest** : Chest circumference in centimeters
- **abdom** : Abdomen circumference in centimeters, measured at the umbilicus and level with the iliac crest
- **hip** : Hip circumference in centimeters
- **thigh** : Thigh circumference in centimeters
- **knee** : Knee circumference in centimeters
- **ankle** : Ankle circumference in centimeters
- **biceps** : Extended biceps circumference in centimeters
- **forearm** : Forearm circumference in centimeters
- **wrist** : Wrist circumference in centimeters, measured distal to the styloid processes

Goal: To predict the **brozek** variable (Percent body fat) using other variables.